

MOROZOVSK, Russian FederationWMO: **345450**Lat: **48.350N**Lon: **41.867E**Elev: **90**StdP: **100.25**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.5	-18.5	-24.7	0.4	-21.1	-21.7	0.5	-18.0	12.2	-2.9	11.1	-2.6	2.2	80	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	35.8	19.1	33.8	18.8	31.8	18.4	21.1	29.5	20.3	28.9	19.6	28.3	4.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	23.5	17.7	12.9	22.8	16.9	12.2	22.4	61.4	29.3	58.8	28.7	56.4	28.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.3	8.1	-24.7	37.5	4.1	2.1	-27.7	39.0	-30.1	40.2	-32.4	41.4	-35.4	42.9		
(5)			-25.0	23.5	4.1	2.2	-27.9	25.1	-30.3	26.4	-32.6	27.7	-35.5	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-5.2	-4.3	1.5	9.6	16.3	20.7	23.6	22.4	16.0	8.8	2.0	-2.9
	DBStd	11.38	6.47	6.49	4.60	4.26	4.45	3.85	3.48	4.07	4.27	5.00	5.29	6.00
	HDD10.0	1931	470	401	266	58	5	0	0	0	4	82	244	401
	HDD18.3	3813	728	634	523	263	94	19	2	10	93	297	491	659
	CDD10.0	1609	0	0	1	45	199	322	422	385	186	45	3	0
	CDD18.3	449	0	0	0	0	30	91	166	137	24	2	0	0
	CDH23.3	5373	0	0	0	19	378	1024	1966	1704	264	18	0	0
	CDH26.7	2372	0	0	0	2	113	419	946	822	67	3	0	0
Wind	WSAvg	3.7	4.1	4.4	4.4	4.1	3.4	3.2	3.0	3.0	3.2	3.2	3.8	4.1
Precipitation	PrecAvg	440	38	30	33	26	45	43	42	28	43	36	34	42
	PrecMax	581	72	85	74	51	146	109	148	72	104	80	86	77
	PrecMin	272	18	7	6	5	1	9	7	3	3	3	5	14
	PrecStd	82	15	18	18	14	32	26	34	19	29	22	19	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	9.4	16.1	26.0	32.6	35.7	37.8	38.8	30.7	25.8	15.8	8.8
		MCWB	5.9	6.8	9.6	14.4	18.1	18.8	19.4	19.2	17.8	16.2	11.8	7.7
	2%	DB	5.1	6.0	12.6	22.1	29.6	33.2	36.0	35.9	28.6	21.4	12.7	7.2
		MCWB	4.3	4.4	7.5	13.3	16.7	18.8	19.3	19.1	16.4	14.1	10.0	6.3
	5%	DB	3.3	4.1	10.2	19.7	27.5	31.2	34.0	33.6	26.5	18.6	10.5	5.6
		MCWB	2.5	3.1	6.5	11.8	16.4	18.5	18.9	18.6	16.1	13.3	8.7	4.7
	10%	DB	2.0	2.7	8.1	17.6	25.2	29.0	31.8	31.3	24.2	16.1	8.8	4.0
		MCWB	1.4	1.8	5.4	10.7	15.8	17.9	18.6	18.1	15.2	12.2	7.5	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.0	10.5	15.8	19.7	21.9	22.2	22.0	20.3	17.6	12.2	8.0
		MCDB	7.3	8.8	14.5	23.8	28.7	29.9	30.9	29.6	27.7	23.7	14.8	8.3
	2%	WB	4.2	4.7	8.4	14.0	18.5	20.6	21.1	20.7	18.6	15.4	10.5	6.3
		MCDB	5.1	5.7	11.4	20.5	26.4	28.6	29.3	29.7	24.8	19.0	12.4	7.1
	5%	WB	2.6	3.2	6.8	12.9	17.4	19.7	20.4	19.9	17.4	14.0	8.9	4.8
		MCDB	3.1	4.0	9.6	18.2	24.8	27.6	29.0	29.0	23.2	17.5	10.2	5.5
	10%	WB	1.4	1.9	5.4	11.7	16.4	18.8	19.7	19.1	16.2	12.6	7.6	3.3
		MCDB	1.8	2.5	7.8	16.3	23.5	26.6	28.5	28.6	21.6	15.8	8.8	3.9
Mean Daily Temperature Range	5% DB	MDBR	5.3	6.0	7.4	11.0	12.0	12.1	12.6	13.5	12.1	9.3	6.2	5.1
		MCDBR	4.3	6.3	11.5	14.4	15.1	15.1	15.2	16.3	15.0	13.0	8.3	5.2
		MCWBR	4.0	5.0	7.3	6.9	6.0	5.3	4.6	5.5	6.3	7.2	6.0	4.6
	5% WB	MCDBR	4.3	5.8	10.2	12.2	12.9	12.5	12.6	13.1	12.3	11.0	7.5	5.1
		MCWBR	4.1	4.8	7.0	6.5	5.8	5.2	4.7	5.5	6.2	6.8	5.8	4.7
Clear-Sky Solar Irradiance	taub		0.333	0.366	0.398	0.449	0.417	0.412	0.438	0.438	0.410	0.377	0.349	0.336
	taud		2.152	2.070	2.130	2.067	2.253	2.302	2.235	2.214	2.261	2.305	2.287	2.192
	Ebn at Noon		695	755	802	798	847	852	822	801	786	748	681	637
	Edh at Noon		85	116	129	152	131	126	133	130	113	92	75	73
All-Sky Solar Radiation	RadAvg		0.98	1.88	3.02	4.48	5.92	6.41	6.20	5.65	4.01	2.29	1.07	0.67
	RadStd		0.23	0.37	0.39	0.44	0.56	0.54	0.34	0.38	0.39	0.31	0.20	0.10